

Deformation flows for the shifted Weil family

The causal generator, an abelian obstruction, and the endpoint direction

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Abstract

A proposal for the bulk-realization program asks whether the shifted Weil family is the deformation flow of a Wilson line: the deformation of a line's expectation obeys a differential equation fixed by the variation of its endpoints together with the non-Abelian Stokes theorem, and the shifted family already obeys a differential equation of that shape. The attraction is that a flow is rigid where a matched value is not. This paper makes the proposal precise and reports what the arithmetic side supplies and what it withholds. We first write the causal generator $G_{\omega,L}$ out in full, which the program's background had defined only by naming its symbol, and show that its three factors deliver the three terms of the target exactly: the shifted form is the $\omega = 0$ form with every off-diagonal kernel and every prime atom multiplied by $\cosh(\omega(x-y))$ and $\cosh(\omega \log n)$, the diagonal unchanged. One consequence is recorded because it is easy to lose: the pole and contact term has identically zero real part on the imaginary axis and reaches the form only through the right half-plane realization. We then prove that truncation to an interval is multiplicative on causal kernels, so that the transfers and their generators lie in one commutative algebra and the path-ordered solution of the flow equation collapses to an ordinary exponential. The shift direction is therefore an abelian line: the mechanism the proposal names has, as things stand, nothing to act on, and any non-commutativity must be supplied transverse to ω . Two positive consequences follow. The shift acts on inputs by the hyperbolic rotation generated by multiplication by x , so it is a geometric deformation and its gauge-theoretic analogue is a cusp angle rather than a coupling. And the endpoint direction, the one the compression supplies, turns out to be an endpoint equation in the exact sense: the L -derivative of the compressed transfer is rank one, sitting at the leading end with the boundary value of the evolved field as its coefficient, so the line has a fixed tail and a moving head. Two things follow. The resulting two-parameter connection is flat, not by hypothesis but because mixed partials commute, so there is no local curvature to extract; and the connection does not close on the form domain, since endpoint evaluation is not form-bounded and the smoothing the shift supplies misses the trace threshold at every admissible shift, by exactly the borderline. What survives is the atomic part of the generator's kernel, whose atoms sit at the prime periods: a flat connection with punctures, whose monodromy is the one place a non-abelian datum could still live, and which is one-dimensional exactly because those residues are scalars. We close with two problems and three necessary conditions a deformation mechanism must satisfy, one of which excludes the displacement operator. Nothing here constructs a source, proves Weil positivity, or assumes the Riemann hypothesis.

1 Introduction

The bulk-realization program of this repository asks for an independently positive system whose state norm is the localized Weil quadratic form [2]. Every candidate tested so far has been asked to

reproduce a *value*: a weight, a pairing, a norm. That has turned out to be a weak demand. In the one setting where the search was carried out concretely, the set of reachable states was everything analytic past the unit circle, so an exhibited weight identity carries no information unless it comes with a reason the representation could not have produced a different weight [5].

This paper takes up a proposal that asks for something else. A Wilson line, when deformed, has an expectation value obeying a differential equation fixed by the variation of its endpoints together with the non-Abelian Stokes theorem. The shifted family of the program already obeys a differential equation of exactly that shape, equation (1.7) of the background [1]:

$$\frac{\partial V_{\omega,L}}{\partial \omega} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I.$$

The proposal is that these are the same equation, so that the object to search for is a *deformation* rather than an operator. The attraction is structural: a generator and an initial condition determine a flow completely, so matching a flow leaves nothing to tune. It is the right kind of demand to make.

What this paper does

One. It writes the generator out. The background defined $G_{\omega,L}$ by naming its symbol and inherited the realization silently from the transfer; Section 2 gives the operator, its domain and its adjoint status, and Section 3 proves that its three factors deliver the three terms of the target exactly. The resulting statement is compact: the shifted form is the central form with every off-diagonal kernel multiplied by $\cosh(\omega(x-y))$ and every prime atom by $\cosh(\omega \log n)$, the diagonal unchanged (Theorem 3.1). The equations of the background are therefore general — they carry the whole Weil form, not the archimedean part of it.

Two. It records a consequence of the causal realization that is easy to lose. The pole and contact term has *identically zero real part on the imaginary axis* (Remark 3.3). Reading the generator symbol as a Fourier multiplier recovers the archimedean term and the prime delays and drops the contact and poles altogether; they reach the form only through the right half-plane Laplace realization.

Three. It proves that the shift direction is abelian. Truncation to an interval is multiplicative on causal kernels (Lemma 4.1), so the transfers and their generators lie in one commutative algebra (Theorem 4.2) and the path-ordered solution of the flow equation collapses to an ordinary exponential. The content of the non-Abelian Stokes theorem is that path ordering may be traded for surface ordering of the parallel-transported curvature; for an abelian connection it degenerates, and a one-parameter family has no surface. *As things stand the mechanism the proposal names has nothing to act on, and the missing non-commutativity has to be supplied transverse to ω .* This is the paper's main result and it is a constraint on the proposal, not a refutation of it.

Four. It identifies what the shift *is*, geometrically. The flow acts on inputs by the hyperbolic rotation generated by multiplication by x (Proposition 5.1), so the shift is a genuine geometric deformation and its gauge-theoretic analogue is an angle — a cusp — rather than a coupling that can be dialled.

Five. It works out the endpoint direction, which is the one the compression supplies and the only one left. Three things come out of it (Section 6). The endpoint variation is *rank one*, sitting at the leading end of the interval with the boundary value of the evolved field as its coefficient, so the line has a fixed tail and a moving head — the endpoint equation of a line, not an analogy for it (Proposition 6.1). The resulting two-parameter connection is *flat*, not by hypothesis but because mixed partial derivatives commute (Proposition 6.2), so there is no local curvature to extract and the naive search for one is misconceived. And the connection does not close on the form domain:

endpoint evaluation is not form-bounded, and the smoothing the shift supplies misses the trace threshold at every admissible shift, by exactly the borderline (Proposition 6.3). What survives is the atomic part of the connection's kernel, whose atoms sit at the prime periods: a flat connection with punctures, whose monodromy is the one place a non-abelian datum could still live. That the monodromy is currently one-dimensional, because the residues are scalars, is what makes the internal index of Remark 4.5 the structural conclusion rather than a free association.

What this paper does not claim

No source is constructed. No positivity is proved. Nothing below assumes the Riemann hypothesis, and no gauge theory is chosen, matched or excluded: the results are statements about the arithmetic side of a proposed correspondence. Section 7 proposes three necessary conditions and labels them as proposals; one of them carries an identification hypothesis that the program deliberately does not make. Proposition 6.3 is a proof sketch resting on a smoothing estimate quoted from the background, and Remark 6.4 is an observation that explains nothing.

Relation to the companion investigations

The conventions, the target and the criterion are those of [2, 3]; Section 2 restates them so that this paper is self-contained, and it is self-contained apart from the background [1], which it cites by equation number for the shifted family, and the companions, which it cites for results quoted by number.

Two inherited facts govern the work and are not re-derived here. The first is that the accumulated exclusions of this program are statements about *channel decompositions* and about the reachable states of particular representations; none of them touches a gauge theory, and none of them constrains a flow, which is not a decomposition [4]. The second is that a mechanism supplying a margin uniform in L proves something false, because the relevant contraction is saturated to more than twenty digits [3]. Section 8 argues that the flow formulation is the one place in the program where that objection may not apply, and states the question rather than answering it.

Finally, the distinction between lines and loops that motivates the whole investigation is not a matter of open versus closed geometry. The test functions carry no boundary conditions; the localization is by support alone. What excluded the closed benchmark was that its operator had bounded spectrum [3, 5], and Section 7 restates that in the form the deformation picture needs.

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2 Conventions, the target, and the shifted family

2.1 The central form

All Hilbert inner products are conjugate-linear in the first argument. For $L > 0$ let $I_L = (-L/2, L/2)$ and let E_L denote extension by zero from I_L to $\mathbb{R}$; unless another domain is stated, $f \in C_c^\infty(I_L; \mathbb{C})$ and $F = E_L f$. The test functions carry no boundary conditions: the localization is by support alone. Write

$$\widehat{F}(\tau) = \int_{\mathbb{R}} e^{-i\tau x} F(x) dx, \quad U_d F(x) = F(x - d), \quad \|F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} |\widehat{F}(\tau)|^2 d\tau, \quad (2.1)$$

and for $r > 0$

$$n_\gamma(r) = \frac{e^{-r/2}}{1 - e^{-2r}} = \sum_{n \geq 0} e^{-a_n r}, \quad a_n = 2n + \frac{1}{2}. \quad (2.2)$$

The archimedean energy, the contact constant and the pole form are

$$K[F] = \int_0^\infty [2\|F\|^2 - 2\operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) dr = \frac{1}{2\pi} \int_{\mathbb{R}} b(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad (2.3)$$

$$b(\tau^2) = \operatorname{Re} \psi(\tfrac{1}{4} + i\tau/2) - \psi(\tfrac{1}{4}), \quad w_0 = \psi(\tfrac{1}{4}) - \log \pi = -5.37218\dots, \quad (2.4)$$

$$P_L[f] = 2 \left| \int_{I_L} f \cosh(x/2) \right|^2 - 2 \left| \int_{I_L} f \sinh(x/2) \right|^2, \quad (2.5)$$

with $\psi = \Gamma'/\Gamma$. The split in (2.4) is bookkeeping: $b(\tau^2) + w_0 = \operatorname{Re} \psi(\tfrac{1}{4} + \frac{i\tau}{2}) - \log \pi$ is the archimedean multiplier, and $\psi(\tfrac{1}{4})$ has been moved between the Dirichlet form and the contact so that $b(0) = 0$. The target is

$$Q_L[f] = K[E_L f] + w_0 \|f\|^2 + P_L[f] - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re}\langle F, U_{m \log p} F \rangle, \quad (2.6)$$

the coefficient being the primitive length $\log p$ and not $m \log p$. Weil's criterion, which we use as input and do not reprove, is

$$\text{RH} \iff Q_L[f] \geq 0 \quad \text{for every } L > 0 \text{ and every } f \in C_c^\infty(I_L). \quad (2.7)$$

Two facts are used repeatedly. Q_L has the spectral form $Q_L[f] = \sum_\rho \widehat{F}(\gamma_\rho) \overline{\widehat{F}(\overline{\gamma_\rho})}$ over the nontrivial zeros, $\gamma_\rho = -i(\rho - \frac{1}{2})$; and the right side is very much smaller than the individual terms of (2.6), which is why assembling Q_L term by term in double precision is unusable [2, 3]. Finally, Q_L is *not* a family of functionals: it is the single Weil functional restricted to $C_c^\infty(I_L)$, since $\langle F, U_r F \rangle = 0$ automatically for $|r| \geq L$. The L -dependence is a support condition on the input and nothing else. This will matter in Section 6.

2.2 The shifted family

Let

$$\xi(s) = \tfrac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \mathcal{F}(p) = \xi(\tfrac{1}{2} + p), \quad (2.8)$$

and for $0 < \omega \leq \frac{1}{2}$ let the causal transfer have Laplace symbol

$$K_\omega(p) = \frac{\mathcal{F}(p - \omega)}{\mathcal{F}(p + \omega)}. \quad (2.9)$$

The causal realization is taken on a line $\operatorname{Re} p = \eta > 1$, where the relevant Dirichlet series converge absolutely and the denominator does not vanish. With P_L multiplication by $\mathbf{1}_{I_L}$, M_η multiplication by $e^{-\eta x}$ and $\mathcal{M}[b]$ the whole-line Fourier multiplier by b ,

$$V_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[K_\omega(\eta + i \cdot)] P_L M_\eta, \quad V_{0,L} = I \text{ (strong limit)}, \quad (2.10)$$

equivalently $V_{\omega,L} f = (k_\omega * \tilde{f})|_{I_L}$ for the inverse Laplace kernel k_ω of K_ω , supported in $[0, \infty)$. This is (1.4) of [1].

Definition 2.1 (The generator). The causal generator symbol is

$$a_\omega(p) = \frac{\mathcal{F}'(p - \omega)}{\mathcal{F}(p - \omega)} + \frac{\mathcal{F}'(p + \omega)}{\mathcal{F}(p + \omega)}, \quad \partial_\omega K_\omega = -a_\omega K_\omega, \quad (2.11)$$

and $G_{\omega,L}$ is realized from a_ω exactly as $V_{\omega,L}$ is realized from K_ω , on the same line and with the same compression:

$$G_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[a_\omega(\eta + i \cdot)] P_L M_\eta, \quad G_{\omega,L} g = (g_\omega * \tilde{g})|_{I_L}, \quad (2.12)$$

with g_ω the inverse Laplace kernel of a_ω , supported in $[0, \infty)$. Its natural domain is $\mathcal{D}_{\log,L} = \{f \in L^2(I_L) : \int \log(2 + |\tau|) |\hat{f}|^2 < \infty\}$, on which the forms below are closed and semibounded.

The operator $G_{\omega,L}$ is not symmetric. Only its symmetric part is a form, and that form is the shifted target:

$$\operatorname{Re} \langle g, G_{\omega,L} g \rangle = Q_{\omega,L}[g], \quad \text{equivalently} \quad G_{\omega,L}^* + G_{\omega,L} \text{ represents } 2Q_{\omega,L}, \quad (2.13)$$

with $Q_{0,L} = Q_L$ of (2.6). Section 3 proves (2.13) by exhibiting the three factors.

2.3 The flow identities

For fixed $f \in C_c^\infty(I_L)$, held fixed as ω varies,

$$\frac{\partial V_{\omega,L}}{\partial \omega} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I, \quad \frac{d}{d\omega} \|V_{\omega,L} f\|^2 = -2 Q_{\omega,L}[V_{\omega,L} f], \quad (2.14)$$

which is (1.7) of [1]. At $\omega = 0$ the first identity is a right derivative on these test functions and no operator-norm derivative is asserted. With the contraction defect $D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}$,

$$\frac{\partial D_{\omega,L}}{\partial \omega} = V_{\omega,L}^* (G_{\omega,L}^* + G_{\omega,L}) V_{\omega,L}, \quad D_{0,L} = 0, \quad (2.15)$$

$$\langle f, D_{\omega,L} f \rangle = 2 \int_0^\omega Q_{s,L}[V_{s,L} f] ds, \quad Q_{0,L}[f] = \lim_{\omega \downarrow 0} \frac{\langle f, D_{\omega,L} f \rangle}{2\omega}, \quad (2.16)$$

which are (1.8) and (1.9) of [1]. Nonnegativity of $D_{\omega,L}$ is exactly $\|V_{\omega,L}\| \leq 1$. The route the deformation picture would have to serve is the following, (Proposition 1.1 of [1]):

Proposition 2.2 (A sufficient contraction route). *If there exist $L_j \rightarrow \infty$ and $0 < \omega_j \downarrow 0$ with $\|V_{\omega_j,L_j}\| \leq 1$ for every j , then the Riemann hypothesis holds.*

We stress two things about this route, because both bear on Section 8. Positivity of $Q_{\omega,L}$ at all shifts is *not* the criterion (2.7); it is a strictly stronger requirement on the family. And the cumulative defect (2.16) can be positive on an input where the instantaneous form is negative, so the two conditions are genuinely different.

3 The generator, and what it contains

Definition 2.1 names an operator; this section identifies it. The outcome is that (2.13) holds with the *complete* target on the right, and that the whole ω -dependence is one scalar factor.

Theorem 3.1 (The shifted form in closed form). *For $0 \leq \omega \leq \frac{1}{2}$ and $f \in C_c^\infty(I_L)$, $F = E_L f$,*

$$\begin{aligned} Q_{\omega,L}[f] = & \int_0^\infty [2\|F\|^2 - 2 \cosh(\omega r) \operatorname{Re}\langle F, U_r F \rangle] n_\gamma(r) dr + w_0 \|f\|^2 \\ & + 2 \iint_{I_L^2} \cosh(\omega(x-y)) \cosh\left(\frac{x-y}{2}\right) \overline{f(x)} f(y) dx dy \\ & - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \cosh(\omega m \log p) \operatorname{Re}\langle F, U_{m \log p} F \rangle. \end{aligned} \quad (3.1)$$

At $\omega = 0$ this is (2.6). In words: the shifted form is the central form with every off-diagonal kernel multiplied by $\cosh(\omega(x-y))$ and every prime atom by $\cosh(\omega \log n)$, the diagonal unchanged.

Proof. Write $\mathcal{F}(p) = \xi(\frac{1}{2} + p)$ and $s = \frac{1}{2} + p$. Logarithmic differentiation of (2.8) gives

$$\frac{\mathcal{F}'}{\mathcal{F}}(p) = \underbrace{\frac{1}{p + \frac{1}{2}} + \frac{1}{p - \frac{1}{2}}}_{\frac{1}{\frac{1}{2}s(s-1)}} \underbrace{-\frac{1}{2} \log \pi + \frac{1}{2} \psi\left(\frac{1}{4} + \frac{p}{2}\right)}_{\pi^{-s/2} \Gamma(s/2)} + \underbrace{\frac{\zeta'}{\zeta}\left(\frac{1}{2} + p\right)}_{\zeta(s)}, \quad (3.2)$$

and $a_\omega(p)$ of (2.11) is the sum of (3.2) at $p - \omega$ and at $p + \omega$. We compute the causal kernel of each factor on $\operatorname{Re} p = \eta > 1$ and then symmetrize. For a causal convolution with kernel c and g supported in I_L ,

$$\operatorname{Re}\langle g, (c * \tilde{g})|_{I_L} \rangle = \frac{1}{2} \iint_{x \neq y} c(|x-y|) \overline{g(x)} g(y) dx dy, \quad (3.3)$$

since relabelling $x \leftrightarrow y$ in the conjugate term turns the region $y < x$ into $x < y$ with $c(|x-y|)$ unchanged.

Pole factors. The four terms $\frac{1}{p \mp \omega \pm \frac{1}{2}}$ have simple poles at $p = \pm \frac{1}{2} \pm \omega$, all to the left of $\operatorname{Re} p = \eta$ since $\eta > 1 \geq \frac{1}{2} + \omega$. Because $\frac{1}{p-a}$ is the Laplace transform of $e^{ax} \mathbf{1}_{x>0}$, their kernel is

$$(e^{x/2} + e^{-x/2})(e^{\omega x} + e^{-\omega x}) \mathbf{1}_{x>0} = 4 \cosh(x/2) \cosh(\omega x) \mathbf{1}_{x>0},$$

and (3.3) turns it into $2 \iint \cosh(\omega(x-y)) \cosh(\frac{x-y}{2}) \overline{f} f$, the third line of (3.1). At $\omega = 0$ the expansion $2 \cosh \frac{x-y}{2} = 2 \cosh \frac{x}{2} \cosh \frac{y}{2} - 2 \sinh \frac{x}{2} \sinh \frac{y}{2}$ returns $P_L[f]$ of (2.5).

Zeta factor. On $\operatorname{Re} p = \eta > 1$ the Dirichlet series converges absolutely, so

$$\frac{\zeta'}{\zeta}\left(\frac{1}{2} + p - \omega\right) + \frac{\zeta'}{\zeta}\left(\frac{1}{2} + p + \omega\right) = -2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) n^{-p},$$

and $n^{-p} = e^{-p \log n}$ is the Laplace symbol of the causal delay by $\log n$. Applying (3.3) to the atom at $r = \log n$ gives $-2 \Lambda(n) n^{-1/2} \cosh(\omega \log n) \operatorname{Re}\langle F, U_{\log n} F \rangle$; writing $n = p^m$, $\Lambda(p^m) = \log p$ gives the last line of (3.1). The cutoff $m \log p < L$ is automatic, because a delay of length $\geq L$ has zero action on I_L .

Gamma and π factors. Their contribution to a_ω is $-\log \pi + \frac{1}{2}[\psi(\frac{1}{4} + \frac{p-\omega}{2}) + \psi(\frac{1}{4} + \frac{p+\omega}{2})]$. With $\psi(z) = -\gamma_E + \int_0^\infty (e^{-t} - e^{-zt})(1 - e^{-t})^{-1} dt$ [8, §5.9] at $z = \frac{1}{4} + \frac{p \pm \omega}{2}$ and the substitution $t = 2r$,

$$\frac{1}{2} \left[\psi\left(\frac{1}{4} + \frac{p-\omega}{2}\right) + \psi\left(\frac{1}{4} + \frac{p+\omega}{2}\right) \right] = -\gamma_E + \int_0^\infty \frac{2e^{-2r}}{1 - e^{-2r}} dr - \int_0^\infty 2 \cosh(\omega r) n_\gamma(r) e^{-pr} dr, \quad (3.4)$$

using $n_\gamma(r) = e^{-r/2}(1 - e^{-2r})^{-1}$ of (2.2). The two integrals in (3.4) diverge logarithmically at the origin and only their difference converges, both integrands being $\sim \frac{1}{2r}$; the pairing of the divergent constant against $\|f\|^2$ with the divergent convolution against $\text{Re}\langle F, U_r F \rangle$ is exactly the combination $2\|F\|^2 - 2\cosh(\omega r) \text{Re}\langle F, U_r F \rangle$ appearing in the first line of (3.1), which is integrable against n_γ because it is $O(r)n_\gamma(r)$ at the origin and $2\|F\|^2 n_\gamma(r)$ for $r \geq L$. This fixes the first line up to an ω -independent constant multiple of $\|f\|^2$, and that constant is $w_0 = \psi(\frac{1}{4}) - \log \pi$ by evaluating at $\omega = 0$, $p = i\tau$, where the multiplier of (2.3) and (2.4) is $\text{Re} \psi(\frac{1}{4} + \frac{i\tau}{2}) - \log \pi = b(\tau^2) + w_0$.

Each of the three contributions is ω -independent on the diagonal, since $\cosh 0 = 1$; summing them gives (3.1), and at $\omega = 0$ gives (2.6). The ω -dependent part of each line is separately convergent, because $\cosh(\omega t) - 1 = O(t^2)$ at the origin. $\square$

Remark 3.2 (The equations of the background are general). Theorem 3.1 settles a question that the background leaves implicit. The flow identities (2.14)–(2.16) are statements about the *whole* Weil form: the archimedean energy, the contact, the poles and the primes all appear, and all are deformed. A separate paragraph of the background isolates the gamma and π factors alone, K_ω^Γ , and observes that a positive tower realizes only that portion of the zero-shift generator; that is a statement about a construction, not about the scope of (2.14).

Remark 3.3 (The pole term is invisible on the imaginary axis). The causal definition on $\text{Re } p = \eta > 1$ is not a technical convenience. At $p = i\tau$,

$$\text{Re} \left[\frac{2}{p - \frac{1}{2}} + \frac{2}{p + \frac{1}{2}} \right] = \frac{2 \cdot \frac{1}{2}}{\frac{1}{4} + \tau^2} + \frac{2 \cdot (-\frac{1}{2})}{\frac{1}{4} + \tau^2} = 0$$

identically in τ . Reading a_ω as a Fourier multiplier on the imaginary axis therefore recovers the archimedean term and the prime delays and *loses the contact and pole term entirely*. The pole factors reach $Q_{\omega,L}$ only through the right half-plane realization of Definition 2.1. Appendix A verifies both halves of this remark numerically: the causal kernel $4\cosh(x/2)\mathbf{1}_{x>0}$ reproduces P_L to machine precision, and the same symbol has vanishing real part on the imaginary axis.

4 The shift direction is abelian

The proposal's mechanism is the non-Abelian Stokes theorem: path ordering along a deformed line may be traded for surface ordering of the parallel-transported curvature, which is what converts a deformation into a differential equation with a local, gauge-invariant right-hand side. This section shows that the mechanism has nothing to act on along ω .

Lemma 4.1 (Truncation is multiplicative on causal kernels). *Let A, B be convolution operators whose kernels are supported in $[0, \infty)$, and let P_L be multiplication by $\mathbf{1}_{I_L}$. Then*

$$P_L A P_L B P_L = P_L A B P_L.$$

Proof. Let f be supported in I_L and put $g = Bf$. Causality of B gives $\text{supp } g \subseteq [-L/2, \infty)$. For $x \in I_L$, causality of A gives $(Ag)(x) = \int_0^\infty a(t)g(x-t)dt$, which involves g only on $(-\infty, x]$, hence only on $[-L/2, x] \subseteq I_L$. Replacing g by $P_L g$ therefore does not change $(Ag)(x)$ for $x \in I_L$, which is the claim. $\square$

Theorem 4.2 (The flow is abelian). *Fix $L > 0$. The operators $\{V_{\omega,L}\}_{0 \leq \omega \leq 1/2}$ and $\{G_{\omega,L}\}_{0 \leq \omega \leq 1/2}$ all lie in one commutative algebra. In particular*

$$[V_{\omega,L}, V_{\omega',L}] = [G_{\omega,L}, G_{\omega',L}] = [G_{\omega,L}, V_{\omega',L}] = 0 \quad \text{for all } \omega, \omega',$$

and the solution of the flow equation (2.14) carries no path ordering:

$$V_{\omega,L} = \exp \left(- \int_0^\omega G_{s,L} ds \right), \quad \text{equivalently} \quad \log K_\omega = - \int_0^\omega a_s ds. \quad (4.1)$$

Proof. By (2.10) and (2.12) every one of these operators has the form $M_\eta^{-1} P_L \mathcal{M}[b(\eta + i \cdot)] P_L M_\eta$ for a symbol b analytic and bounded on $\text{Re } p = \eta$, and the conjugation by M_η replaces a kernel $k(x - y)$ by $k(x - y)e^{\eta(x-y)}$, again a causal function of $x - y$. So each is the truncation to I_L of a convolution with kernel supported in $[0, \infty)$. By Lemma 4.1 the truncation map is multiplicative on this class, hence an algebra homomorphism onto its image, and convolution is commutative. This gives the three vanishing commutators.

A commuting family of generators makes the path-ordered solution of $\partial_\omega V = -GV$, $V_0 = I$, the ordinary exponential of $-\int_0^\omega G_{s,L} ds$; equivalently, applying the homomorphism to the whole-line identity $\partial_\omega K_\omega = -a_\omega K_\omega$, $K_0 = 1$, which integrates to the scalar statement on the right of (4.1). $\square$

Remark 4.3 (Discrete form). In a discretization ordered by x , Lemma 4.1 is the remark that a truncated causal convolution is a lower-triangular Toeplitz matrix, hence a power series in one nilpotent shift, and that any two such commute. Appendix A records this as an exact rational computation, together with the control showing that causality is what does the work: truncated *non-causal* convolutions do not commute.

Corollary 4.4 (No non-Abelian Stokes content along the shift). *The shift flow is the holonomy of an abelian connection on a one-dimensional parameter interval. Its holonomy depends on the endpoints of that interval and on nothing else; there is no surface for a curvature to be integrated over, and the non-Abelian Stokes theorem degenerates to the fundamental theorem of calculus, which is the right-hand identity in (4.1).*

Remark 4.5 (What this costs the proposal, and what it asks for). Corollary 4.4 is not a refutation. It is a statement of what is missing, and it is precise: *any non-commutativity must be supplied transverse to ω* . There are exactly three places it could come from within the objects this program already has.

1. **The endpoint direction.** P_L itself is not in the algebra of Theorem 4.2; the nested projections commute with one another but not with the transfers. This is Section 6, and it is the direction the proposal itself names.
2. **The bilinear energy balance.** The second identity of (2.14) is quadratic in $V_{\omega,L}$ where the first is linear: $\partial_\omega \|Vf\|^2 = -\langle Vf, (G^* + G)Vf \rangle$ cuts the line at the deformation point, inserts an object and pairs the two halves. That is the shape of the loop equation [10], whose right-hand side is likewise bilinear in the two segments meeting at the deformation point, and in that correspondence the Weil form occupies the slot the parallel-transported curvature occupies. The caution is that the loop equation is an equation for an expectation and is nonlinear in the loop functional, whereas (2.14) is linear at the operator level; a matching attempt must say which of the two it claims.
3. **An internal index.** The prime coefficients of (3.1) are scalars. The corresponding sum for an Artin or automorphic L -function carries $\text{tr } \rho(\text{Frob}_p)$, a trace of a holonomy in a representation, which is the one place in this subject where a non-abelian holonomy is not an analogy. We record this and do not pursue it: it changes the target from ζ to a family and nothing in the investigation so far requires it.

Remark 4.6 (Relation to the two-dimensional control). An earlier calculation in this program used rigorously constructed two-dimensional Yang–Mills [9] to reject literal Wilson-loop winding as the prime repetition law, on the ground that winding is not transfer evolution [6]. Theorem 4.2 is a statement about the transfer evolution itself and is independent of that calculation; the two are complementary, the first saying what the evolution is not and the second saying what structure the evolution does not have.

5 The shift is a hyperbolic rotation

Theorem 3.1 has an immediate consequence that identifies the shift geometrically. Let X be multiplication by x , so that $\cosh(\omega X)$ and $\sinh(\omega X)$ map $C_c^\infty(I_L)$ to itself.

Proposition 5.1 (Hyperbolic form of the shift). *For $0 \leq \omega \leq \frac{1}{2}$ and $f \in C_c^\infty(I_L)$,*

$$Q_{\omega,L}[f] = Q_{0,L}[\cosh(\omega X)f] - Q_{0,L}[\sinh(\omega X)f]. \quad (5.1)$$

Proof. Write $g_c = \cosh(\omega X)f$, $g_s = \sinh(\omega X)f$, $G_c = E_L g_c$, $G_s = E_L g_s$, and evaluate (2.6) on each, term by term. For the norms, $|\cosh(\omega x)|^2 - |\sinh(\omega x)|^2 = 1$ pointwise, so $\|g_c\|^2 - \|g_s\|^2 = \|f\|^2$; this disposes of the contact term and of the $2\|F\|^2$ inside (2.3). For a translation, at any r ,

$$\begin{aligned} \operatorname{Re}\langle G_c, U_r G_c \rangle - \operatorname{Re}\langle G_s, U_r G_s \rangle &= \operatorname{Re} \int [\cosh(\omega x) \cosh(\omega(x-r)) \\ &\quad - \sinh(\omega x) \sinh(\omega(x-r))] \overline{F(x)} F(x-r) dx \\ &= \cosh(\omega r) \operatorname{Re}\langle F, U_r F \rangle, \end{aligned}$$

by the addition formula $\cosh a \cosh b - \sinh a \sinh b = \cosh(a-b)$. Applying this at r inside the n_γ integral and at $r = m \log p$ in the prime sum reproduces the first and last lines of (3.1). For the pole form, $P_L[g] = 2 \iint \cosh(\frac{x-y}{2}) \overline{g(x)} g(y)$, and the same addition formula applied to $\cosh(\omega x) \cosh(\omega y) - \sinh(\omega x) \sinh(\omega y) = \cosh(\omega(x-y))$ reproduces the third line. Each difference is computed inside a convergent integral, so no regularization is needed. Summing gives (3.1), which is $Q_{\omega,L}[f]$ by Theorem 3.1. $\square$

Equation (5.1) is (1.10) of [1]; the proof above derives it from the generator rather than from a kernel expansion, which is what makes the next remark available.

Remark 5.2 (The shift is an angle, not a coupling). The pair $(\cosh(\omega X), \sinh(\omega X))$ is a hyperbolic rotation by angle ω with generator X . Two consequences are worth stating.

First, the shift is a *geometric* deformation of the input, not a change of coefficients, so it is the kind of parameter a line can carry. In gauge theory the natural object with a hyperbolic angle attached to a line is a *cusp*; this is where we would look first among specific theories, and the differential equation a cusped line's expectation satisfies in its angle has exactly the shape the proposal asks for [11].

Second, the parameter is not free. A cusp angle is a geometric datum of the contour, so a mechanism built on it cannot be dialled to fit; that is the property the program wants, since any mechanism with a tunable margin uniform in L proves something false [3]. The obstruction to state plainly is one of type: a cusp anomalous dimension is a number, while $Q_{\omega,L}$ is a form on an infinite-dimensional input space, so a line carrying it must carry an infinite-dimensional label. This is Condition 7.1 below.

Remark 5.3 (Positivity at positive shift is a different statement). Proposition 5.1 exhibits $Q_{\omega,L}$ as a *difference* of two central energies. Nonnegativity of each does not give nonnegativity of the difference, so positivity of the shifted form at all shifts is strictly stronger than the criterion (2.7) and is not implied by the Riemann hypothesis. The route of Proposition 2.2 does not require it: what it requires is $\|V_{\omega_j,L_j}\| \leq 1$, a statement about the cumulative defect (2.16) rather than about its integrand.

6 The endpoint direction

Theorem 4.2 sends us transverse to ω , and the proposal's own phrase — the variation of the endpoints — names the direction. The compression P_L is not in the commutative algebra: the projections commute with one another, but not with the transfers. This section works out what the L -direction is, and reaches three conclusions, of which the third is the one to carry forward.

6.1 The variation is rank one, at the leading endpoint

Let V_ω denote the uncompressed causal transfer, so that $V_{\omega,L}f = P_L V_\omega f$ for f supported in I_L . Fix $f \in C_c^\infty(I_{L_0})$ and let $L > L_0$, so that f is held fixed while the interval grows.

Proposition 6.1 (Endpoint variation). *For $L > L_0$ and $f \in C_c^\infty(I_{L_0})$,*

$$\frac{\partial}{\partial L}(V_{\omega,L}f) = \frac{1}{2} (V_\omega f)(L/2) \delta_{L/2}, \quad (6.1)$$

a rank-one variation supported at the single point $x = L/2$, whose coefficient is the boundary value of the evolved field. Equivalently $\partial_L V_{\omega,L} = B_L V_{\omega,L}$ with $B_L = \frac{1}{2} \delta_{L/2} \otimes \text{ev}_{L/2}$.

Proof. $V_{\omega,L}f = \mathbf{1}_{I_L} \cdot (V_\omega f)$ and $\partial_L \mathbf{1}_{(-L/2,L/2)} = \frac{1}{2}(\delta_{L/2} + \delta_{-L/2})$, so the variation has the two endpoint terms. Causality kills the second: f is supported in I_{L_0} and the kernel of V_ω is supported in $[0, \infty)$, so $\text{supp } V_\omega f \subseteq [-L_0/2, \infty)$ and $(V_\omega f)(-L/2) = 0$ for $L > L_0$. The surviving coefficient is the boundary value of $V_{\omega,L}f$ from inside, since $V_\omega f$ and $V_{\omega,L}f$ agree on I_L . $\square$

Equation (6.1) is exactly the endpoint equation of a line: the variation is local, it sits at the moving end, and its coefficient is the transported field there. The tail is fixed and the head moves, which is what causality means here. This is the closest structural match the proposal has, and it is not an analogy: the two statements have the same form.

6.2 The two-parameter connection is flat, so there is no curvature to find

It is tempting to expect a curvature in the (ω, L) plane and to look for it in the mixed derivative. That expectation is wrong, and it is worth saying why, because the correction is what sends the search somewhere else.

Proposition 6.2 (Flatness). *On the range of the two-parameter family, wherever $V_{\omega,L}f$ is twice continuously differentiable in (ω, L) ,*

$$\partial_\omega B_L + \partial_L G_{\omega,L} + [G_{\omega,L}, B_L] = 0. \quad (6.2)$$

Proof. Mixed partial derivatives of $V_{\omega,L}f$ commute. Substituting $\partial_\omega V_{\omega,L} = -G_{\omega,L} V_{\omega,L}$ from (2.14) and $\partial_L V_{\omega,L} = B_L V_{\omega,L}$ from Proposition 6.1 into $\partial_\omega \partial_L = \partial_L \partial_\omega$ and collecting terms gives (6.2) applied to $V_{\omega,L}f$. $\square$

So the pair $(-G_{\omega,L} d\omega, B_L dL)$ is a *flat* connection wherever it is a connection at all: the zero-curvature condition is not an extra requirement to be imposed, it is forced by the existence of the two-parameter family. There is no local curvature to compute and no surface term to extract, and the non-Abelian Stokes theorem still has nothing to integrate. On a simply connected parameter domain, the holonomy of a flat connection depends on the endpoints alone.

6.3 The connection does not close, and it misses by exactly the borderline

Proposition 6.1 holds on the core $C_c^\infty(I_L)$, where $V_\omega f$ is smooth because $\hat{f}$ decays rapidly. It does not extend to the form domain.

Proposition 6.3 (No form-bounded endpoint evaluation). *B_L is not bounded relative to $Q_{0,L}$, at any $0 \leq \omega \leq \frac{1}{2}$.*

Proof sketch. B_L is built from $\text{ev}_{L/2}$, and a point evaluation on $\mathbb{R}$ is bounded on H^r only for $r > \frac{1}{2}$. The form domain is $\mathcal{D}_{\log,L}$ of Definition 2.1, whose multiplier $b(\tau^2) = \log |\tau| + O(1)$ is only logarithmically stronger than L^2 ; it is contained in no H^r with $r > 0$ and carries no trace operator, which is also why the natural closed domain of the archimedean energy has no trace condition [2]. The transfer does not repair this: the bound $|K_\omega(\eta + i\tau)| \leq C_\eta(2 + |\tau|)^{-\omega}$ of [1] gives V_ω a smoothing gain into H^r for $0 < r < \min(\omega, \frac{1}{2})$ only, which is strictly below the trace threshold for every admissible shift, the endpoint $\omega = \frac{1}{2}$ included. $\square$

Remark 6.4 (A third marginality at the same exponent). The failure in Proposition 6.3 is not by a wide margin; it is exactly borderline. Three separate quantities in this program sit at the same exponent and on the wrong side of it: the archimedean kernel $n_\gamma(r) \sim \frac{1}{2r}$ makes K the Gagliardo seminorm at $s = 0$, the borderline of the scale, so the form domain is only logarithmically better than L^2 ; the transfer gains at most $(2 + |\tau|)^{-1/2}$; and a trace needs $r > \frac{1}{2}$. We record the coincidence and do not explain it. Whether it is one fact seen three times is Problem 6.5.

6.4 The primes are the atoms of the connection

The proof of Theorem 3.1 decomposes the kernel of $G_{\omega,L}$ into a continuous part and an atomic part:

$$\begin{aligned} g_\omega(t) &= g_\omega^{\text{cont}}(t) + g_\omega^{\text{at}}(t), \quad t > 0, \\ g_\omega^{\text{cont}}(t) &= 4 \cosh(\omega t) \cosh(t/2) - 2 \cosh(\omega t) n_\gamma(t) + \text{const}, \\ g_\omega^{\text{at}}(t) &= -2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) \delta_{\log n}, \end{aligned} \tag{6.3}$$

the atoms sitting at the prime periods $m \log p$ with masses $2(\log p) p^{-m/2} \cosh(\omega m \log p)$. This is the Lévy decomposition that the companion investigation's ninth selection rule requires of any source [3], and it is here a property of the connection rather than of a prepared state.

Read along the line, an atomic connection is a connection with *punctures*: the holonomy of the segment is an ordered product of one local factor per atom crossed, times the contribution of the continuous part. That is precisely the periodic-orbit structure the program's own analysis says the primes demand and which a short-distance mechanism cannot supply [5]. It is also, given Propositions 6.2 and 6.3, the only place left for the proposal's mechanism to live: a flat connection has no local curvature, but a flat connection on a punctured domain has monodromy, and monodromy is exactly the non-local, potentially non-abelian datum the proposal needs.

The obstruction is stated by Theorem 4.2. The local factors at different atoms commute, because every object here is a convolution and the atomic masses are *scalars*. The monodromy representation

is therefore one-dimensional and carries no more information than the product of its masses. To make it carry more, the residues at the atoms would have to be matrix valued — and the arithmetic object whose prime sum carries matrix-valued residues is $\text{tr } \rho(\text{Frob}_p)$ for a representation ρ . The three strands of Remark 4.5 converge on this point: the endpoint direction is where the geometry is, the atoms are where the primes are, and an internal index at the atoms is the only way to make the two non-abelian at once. We state this as the structural conclusion of the paper and not as a programme, since it changes the target from ζ to a family.

6.5 Two open problems

Problem 6.5 (Close the endpoint flow, or show it cannot be closed). Proposition 6.3 obstructs B_L on the form domain while Proposition 6.1 defines it on a dense core. Determine whether the endpoint flow admits a closed realization on any domain on which $Q_{\omega,L}$ is also closed — for instance by pairing the endpoint evaluation against the atomic part of (6.3), which is where the flow’s own data sits — or prove that the borderline failure of Remark 6.4 is structural.

Problem 6.6 (Monodromy of the punctured flow). Compute the local factor of (6.3) at a single atom, as an operator on the core, and determine what a matrix-valued residue at that atom would have to satisfy for the resulting holonomy to remain a contraction in the sense of Proposition 2.2. This is the smallest calculation that would tell whether the internal index of Remark 4.5(3) can be introduced without destroying the flow identities.

Remark 6.7 (What the L -direction is not). Two cautions, both of which this program has paid for once. First, Q_L is not a family of functionals but one functional restricted by support, so a realization at fixed L carries no information and an L -family of unrelated realizations proves nothing [2]. The L -dependence that is real is in the compression, which is what Proposition 6.1 differentiates. Second, the exclusions this program has accumulated are statements about channel decompositions; a flow is not a decomposition, so they neither block this direction nor may be quoted in its support [4].

7 Conditions on a deformation mechanism

A candidate arising from this picture inherits the necessary conditions the companion investigation has accumulated [3]. We do not restate their derivations; the table records which ones bear on a flow and how.

Inherited condition	Bearing on a deformation mechanism
Reflection invariance	The flow must commute with $x \mapsto -x$; free to check, and the involution is the functional equation acting on the zeros.
Unboundedness relative to L^2	$Q_L[\chi e^{iNx}]/\log N \rightarrow \ \chi\ ^2$, so no bounded preparation matches. This is what excluded the closed benchmark; Condition 7.1 restates it for a holonomy.
Spectral weight and channel tower	The archimedean multiplier is the smooth zero density to four orders, with the exact tower $\{2n + \frac{1}{2}\}$ and unit residues. Condition 7.3 reads this as a defect two-point function.
Criticality	The relevant contraction is saturated and the margin collapses superexponentially in L ; any mechanism supplying a margin uniform in L proves something false. See Section 8, where we ask whether the flow formulation is the exception.
Jump structure	The non-pole part of the target is one Lévy jump form with atoms at $m \log p$. Equation (6.3) realizes this on the connection.

We add three conditions specific to the deformation picture. They are *proposed*: each is elementary, none has been tested against a specific theory, and the third carries an identification hypothesis that this program deliberately does not make.

Condition 7.1 (Unbounded connection, infinite-dimensional representation). A holonomy valued in a compact group, in a finite-dimensional representation, is unitary; its matrix elements and its character are bounded functions, so every pairing assembled from them is bounded on $L^2(I_L)$ and is excluded by the unboundedness rule. A deformation mechanism must therefore carry an unbounded connection in an infinite-dimensional representation.

This is the structural form of an exclusion the program already had. The closed benchmark failed because its operator acted as $W = 2 \cos \theta$, with spectrum in $[-2, 2]$, so that its kernel was entire in the separation and had no atoms [3]. Condition 7.1 says that the boundedness was not an accident of that benchmark: $2 \cos \theta$ is the character of a holonomy in the fundamental of $SU(2)$, and *every* such character is bounded. Note also that $V_{\omega,L}$ is not unitary — it is a contraction, and the object of interest is its defect (2.15) — so the connection to be matched is not anti-Hermitian either.

Condition 7.2 (Reflection and the atoms). The connection's kernel must be a continuous part plus atoms at the prime periods $m \log p$ with masses $2(\log p)p^{-m/2} \cosh(\omega m \log p)$, and the whole must be even in the separation after symmetrization. A connection that is smooth in the separation cannot produce the atoms, and a connection whose atoms sit at other positions is excluded without any positivity computation.

Condition 7.3 (Marginal short distance: effective dimension $\frac{1}{2}$). Suppose the line's parameter is identified with the arithmetic coordinate x . Since $n_\gamma(r) = \frac{1}{2r} + O(1)$ as $r \downarrow 0$ by (2.2), (2.3) gives

$$K[F] = \frac{1}{2} \iint |F(x) - F(y)|^2 n_\gamma(|x - y|) dx dy \sim \frac{1}{4} \iint \frac{|F(x) - F(y)|^2}{|x - y|} dx dy$$

near the diagonal: the Gagliardo seminorm at the borderline exponent $1 + 2s = 1$, that is $s = 0$. An insertion on the line reproducing this must therefore have two-point function $|x - y|^{-1}$, an effective defect dimension $\Delta = \frac{1}{2}$, with the coefficient fixed rather than free.

Condition 7.3 has one immediate consequence, and it excludes the first thing one would try. The displacement operator of a line defect has protected dimension $\Delta = 2$, fixed by the Ward identity that defines it, and the standard protected scalar insertions on a supersymmetric Wilson line have $\Delta = 1$ [12, 11]. Neither is $\frac{1}{2}$. *So the deformation whose generator matches is not the displacement operator*, which is what a deformation of the contour would naively produce.

Remark 7.4 (The identification hypothesis). Condition 7.3 assumes the line's parameter is the arithmetic coordinate. The program does not make that identification: the arithmetic coordinate is not automatically a physical Euclidean time or contour parameter, and an earlier calculation excluded one natural way of making it one [6, 2]. Sections 4 and 6 suggest a different assignment, in which ω is the parameter along the line — a cusp angle, by Remark 5.2 — and x indexes the representation the line carries. Under that assignment Condition 7.3 moves and must be rederived, while Conditions 7.1 and 7.2 are unaffected. Deciding between the two assignments is the first question a specific proposal has to answer.

8 Outlook

8.1 What the paper leaves

Section 3 is settled: the generator is written out, the three factors deliver the three terms of the target, and the whole shift dependence is one cosh factor multiplying the off-diagonal data. Sections 4 and 6 narrow the proposal rather than advancing it. The shift direction is abelian; the two-parameter connection is flat wherever it exists; the endpoint connection does not close on the form domain, and misses by exactly the borderline. What survives is the atomic structure (6.3): a flat connection with punctures at the prime periods, whose monodromy is currently one-dimensional because the residues are scalars. Problems 6.5 and 6.6 are the two smallest calculations that would move this.

8.2 The decisive question

The program's sharpest structural constraint is that the contraction whose norm is Weil positivity is saturated, with a margin collapsing superexponentially in L , so that any mechanism supplying a margin uniform in L proves something false [3]. That statement is about the central form at $\omega = 0$. Proposition 2.2 asks for something else: $\|V_{\omega_j, L_j}\| \leq 1$ along a sequence with $\omega_j \downarrow 0$, which by (2.16) is a statement about the cumulative defect $\langle f, D_{\omega, L} f \rangle = 2 \int_0^\omega Q_{s, L} [V_{s, L} f] ds$, not about its integrand. The two differ: the cumulative defect can be positive on an input where the instantaneous form is negative, and the program's own records contain such a case at $L = \log 7$, $\omega = 10^{-11}$ [1].

Problem 8.1 (Does the flow formulation escape the criticality?). Decide whether a strict contraction $\|V_{\omega, L}\| \leq 1 - \varepsilon$ at fixed $\omega > 0$, uniform in L , is consistent — with the criterion, with the explicit formula, and with the saturation at $\omega = 0$. If it is, then the flow formulation is the only formulation in this program in which a mechanism with a margin is not immediately self-refuting, and that is the strongest available argument for the whole direction. If it is not, the deformation picture inherits the criticality unchanged and offers no relief from it.

This should be settled before a gauge theory is chosen, not after. It is a question about the arithmetic side alone.

8.3 Order of work

1. Problem 8.1. Cheapest to state, decisive for the direction, and independent of any choice of theory.
2. Problem 6.6: the local factor at one atom, and what a matrix-valued residue would have to satisfy.
3. Problem 6.5: whether the endpoint flow closes anywhere.
4. Only then, a specific theory. Remark 5.2 says where to look first — a cusp, because the shift is an angle — and Section 7 is the battery to run before computing anything about a candidate. Condition 7.3 already rejects the obvious first attempt.

8.4 Scope

Nothing above is proposed as a route to the Riemann hypothesis. Theorem 3.1, Lemma 4.1, Theorem 4.2, Proposition 5.1, Proposition 6.1 and Proposition 6.2 are written proofs; Proposition 6.3 is a proof sketch, resting on a smoothing estimate quoted from [1] and on the standard Sobolev trace threshold. Remark 6.4 is an observation about three exponents and explains nothing. Section 7’s three conditions are proposals and are not part of the companion investigation’s index. No source is constructed, no positivity is proved, no gauge theory is matched or excluded, and no statement here assumes the Riemann hypothesis.

Preparation

This working manuscript was produced with a large language model, Anthropic’s Claude (model `claude-opus-5`), working directly in the investigation repository: the derivations and proofs of Sections 3–6, the conditions of Section 7, the check program of Appendix A, and the drafting throughout. The proposal the paper analyses is Edward Baker’s. The research notes accompanying the investigation name the model used for each. Every statement is classified in the text as a written proof, a proof sketch, a labelled numerical computation, or an assessment.

A Checks

One program,

```
numerics/check_causal_commutation.py,  
numerics/records/causal-commutation-checks.json,
```

the second being its preserved record. It uses the standard library only and prints JSON to standard output. Exact rational arithmetic is labelled separately from floating-point work. Counts refer to finite test cases, not to independent theorems, and none of the checks is a positivity certificate.

Group	What is verified
Multiplicativity of truncation	Lemma 4.1 as a finite matrix identity, $P_L A P_L B P_L = P_L A B P_L$, in exact rational arithmetic, for randomly generated causal kernels at several truncation sizes.
Commutation, and its cause	$[A, B] = 0$ for truncated causal convolutions, in exact rational arithmetic (Theorem 4.2, Remark 4.3); and the control that truncated <i>non</i> -causal convolutions have commutators bounded away from zero, so that causality is what the lemma uses.
The causal pole kernel	That $4 \cosh(\omega x) \cosh(x/2) \mathbf{1}_{x>0}$, symmetrized by (3.3), equals $2 \cosh(\omega(x-y)) \cosh(\frac{x-y}{2})$, and at $\omega = 0$ equals P_L of (2.5) in the form $2 C(f) ^2 - 2 S(f) ^2$ (Theorem 3.1, Remark 3.3).
Vanishing on the imaginary axis	That $\operatorname{Re}[2(p - \frac{1}{2})^{-1} + 2(p + \frac{1}{2})^{-1}] = 0$ at $p = i\tau$, exactly, over a grid of τ (Remark 3.3).
The hyperbolic identity	Proposition 5.1 as a finite matrix identity: the $\cosh(\omega(x-y))$ -weighted form equals $Q_{0,L}[\cosh(\omega X)f] - Q_{0,L}[\sinh(\omega X)f]$ on a discretized kernel, at several ω and L .

Run from the investigation directory:

```
python3 numerics/check_causal_commutation.py
python3 validation/drafts.py check --replay
```

The checks verify finite algebra underlying the identities of Sections 3–5. They do not verify the convergence arguments in the proof of Theorem 3.1, which are analytic, nor Proposition 6.3, which is a statement about domains.

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